

Difference Between JPA, Hibernate, and Spring Data JPA

JPA (Java Persistence API)

JPA is a **Java specification** that provides a standard way to map Java classes to database tables. It defines annotations such as **@Entity**, **@Table**, **@Id**, and **@GeneratedValue**. Since JPA is only a specification, it requires an implementation such as **Hibernate** to perform database operations.

Example:

```
@Entity
@Table(name="employees")
public class Employee {
    @Id
    @GeneratedValue(strategy=GenerationType.IDENTITY)
    private Long empId;
    private String empName;
}
```

Hibernate

Hibernate is an ORM framework that implements JPA. It converts Java objects into database records and automatically generates SQL queries.

Example:

```
Employee employee = new Employee();
employee.setEmpName("Anjali");
session.persist(employee);
```

Spring Data JPA

Spring Data JPA is built on top of JPA and provides repository interfaces like **JpaRepository** to perform CRUD operations with minimal code.

Example:

```
public interface EmployeeRepository extends JpaRepository<Employee, Long> {}
employeeRepository.save(employee);
employeeRepository.findById(1L);
```

Relationship

JPA defines the standard, Hibernate implements it, and Spring Data JPA simplifies development using built-in repository methods.

In Simple Terms: JPA – Defines the rules. Hibernate – Implements the rules. Spring Data JPA – Simplifies database operations.

Conclusion

JPA provides the specification, Hibernate performs the persistence operations, and Spring Data JPA reduces boilerplate code, making application development faster and easier.